

# Slides for the MFAI (Jul-Dec 2026) Lectures

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# Artificial Intelligence

- ▶ Artificial Intelligence - refers to the development of computer systems capable of performing tasks that typically require human intelligence, such as learning, problem-solving, and decision-making. It encompasses technologies that enable machines to mimic human-like cognitive functions (Google).
- ▶ AI is focused on developing and studying methods and software that enable machines to *perceive their environment* and use *learning* and *intelligence* to take actions that maximise their chances of achieving defined goals (from Google).
- ▶ Goals : learning, reasoning, knowledge representation, planning, natural language processing, perception, etc.
- ▶ Techniques employed : Search, mathematical optimisation, formal logic, artificial neural networks, also drawing upon methods from statistics, economics, operations research, etc.

# Machine Learning

- ▶ Machine Learning is a field of study focused on developing and studying statistical algorithms that can learn from data and *generalise* to unseen data, and performs tasks without explicit instructions (Google).
- ▶ ML is applied in natural language processing, computer vision, speech recognition, email filtering, agriculture, medicine and predictive analytics (from Google).
- ▶ Statistics and mathematical optimisation form the foundation of ML.
- ▶ Probably Approximately Correct (PAC) learning - theoretical framework for machine learning.

# Linear Algebra

- ▶ Algebra of linear mappings over real/complex vectors and matrices. More generally, algebra of linear mappings between vector spaces.
- ▶ vectors :  $x, y \in R^n : x = (x_i)_i = (x_1, \dots, x_n), y = (y_i)_i$ .
- ▶ special case : matrices :  $A \in R^{m \times n} ; A = (a_{ij})_{i,j}, i \in [m], j \in [n]$ .
- ▶ vector addition :  $x + y = (x_i)_i + (y_i)_i = (x_i + y_i)_{i=1}^n ;$
- ▶ scalar multiplication of a vector :  $\alpha x = (\alpha x_i)_i, \alpha \in R$ .
- ▶ matrix addition :  $A + B = (a_{ij})_{ij} + (b_{ij})_{ij} = (a_{ij} + b_{ij})_{ij} ;$
- ▶ scalar multiplication of a matrix :  $\alpha A = \alpha (a_{ij})_{i,j} = (\alpha a_{ij})_{i,j}$ .
- ▶ Matrix multiplication :  $A \in R^{m \times n}, B \in R^{n \times r}$
- ▶ Define  $AB \in R^{m \times r}$  by  $(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$  for all  $i, j$ .
- ▶ In particular,  $A \in R^{m \times n}, x \in R^n \Rightarrow Ax \in R^m$ .
- ▶ matrices - why study them ? One Reason : System of linear equations :
- ▶  $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1,$
- ▶  $a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2, \dots,$
- ▶  $a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m.$
- ▶ It can be expressed as  $Ax = b$  where  $A \in R^{m \times n}, x \in R^n, b \in R^m$ .

# Vector Spaces

- ▶  $V$  = vectors ;  $F$  - field ;  $+$  :  $V \times V \rightarrow V$  ;  $\cdot$  :  $F \times V \rightarrow V$ .
- ▶  $+$  is the vector addition ;  $\cdot$  is the scalar multiplication. 1 is the multiplicative identity of  $F$ .
- ▶  $(V, F, +, \cdot, 1)$  forms a vector space (VS) if
- ▶ 1)  $(V, +)$  forms an abelian group.
- ▶ 2)  $\cdot$  distributes over  $+$  :  $a \cdot (u + v) = a \cdot u + a \cdot v, \forall a \in F, u, v \in V$ .
- ▶ 3)  $a \cdot (b \cdot u) = (ab) \cdot u, \forall a, b \in F, u \in V$  ; 4)  $1 \cdot u = u, \forall u \in V$ .
- ▶
- ▶ Euclidean spaces  $R^2, R^3$  are vector spaces over  $R$ .
- ▶  $V = F^n, W = F^{m \times n}$  are vector spaces over  $F$  ;
- ▶  $V$  = set of all polynomials (of degree at most 4) with coefficients from  $F$ , is a vector space over  $F$ .
- ▶ We set  $F = R$  and  $V = R^n$  or  $R^{m \times n}$  for this course.

# Vector Subspaces

- ▶ **Defn :** For  $U \subseteq V$ ,  $U \neq \emptyset$ , if  $(U, F, +, \cdot, 1)$  itself forms a VS, then  $U$  is a subspace of  $V$ .
- ▶ **Theorem :**  $U \subseteq V$  forms a subspace of  $(V, F, +, \cdot, 1)$  if and only if  $au + bv \in U$  for all  $u, v \in U$ ,  $a, b \in F$ .
- ▶ Examples of subspaces : lines  $\{ax + by = 0\}$  and planes  $\{ax + by + cz = 0\}$  passing through the origin.
- ▶  $A \in R^{m \times n} \Rightarrow \{x \in R^n : Ax = 0\}$  is a subspace of  $R^n$ . Referred to as Null Space of  $A$  and denoted by  $N(A)$ .
- ▶  $A \in R^{m \times n} \Rightarrow \{Ax : x \in R^n\}$  is a subspace of  $R^m$ . Referred to as the Range Space of  $A$  and denoted by  $R(A)$ .
- ▶  $U, W \subseteq V$  - subspaces of  $V$ .  $U \cap W$  is a subspace of  $V$ .  $U + W := \{u + w : u \in U, w \in W\}$  is a subspace of  $V$ .
- ▶ intersection of any collection of subspaces is also a subspace. Addition of any finite collection of subspaces is a subspace.

# Vector Subspaces

- ▶ **Defn** :  $V$  is a direct sum,  $V = U \oplus W$  of subspaces  $U$  and  $W$  if  $V = U + W$  and  $U \cap W = \{0\}$ .
- ▶ Example :  $R^3$  is a direct sum of  $U =$  the x-y plane  $\{(x, y, 0)\}_{x,y}$  and the z axis  $\{(0, 0, z)\}_z$ .
- ▶  $V = U \oplus W \iff$  each  $v \in V$  has a unique representation  $v = u + w$  for some  $u \in U, w \in W$ ,
- ▶ **Defn** :  $S = \{u_1, \dots, u_n\}$  and  $u : u$  is a *linear combination (l.c.)* of  $S$  if, for some  $a_1, \dots, a_n \in F$ , we have  $u = a_1 u_1 + \dots + a_n u_n$ .
- ▶ Eg : For any  $x = (x_1, x_2, x_3)$ ,  $x = x_1 e_1 + x_2 e_2 + x_3 e_3$ .
- ▶  $\forall (x, y) \in R^2$ ,  $(x, y) = \sqrt{2}y(\cos 45, \sin 45) + (x - y)(1, 0)$ .
- ▶ If  $u$  is a l.c. of  $S = \{u_1, \dots, u_n\}$ , then  $u$  is *linearly dependent* on  $S$ . Else, it is *linearly independent* of  $S$ .
- ▶  $\text{span}(S) = \{a_1 u_1 + \dots + a_n u_n : n \geq 0, u_1, \dots, u_n \in S, a_1, \dots, a_n \in F\}$ .  $\text{span}(S)$  is *generated* by  $S$ .
- ▶ **Theorem** :  $\text{span}(S)$  is a subspace of  $V$  for every  $S \neq \emptyset$ .

# Linear independence

- ▶ **Defn** : A finite set  $S = \{u_1, \dots, u_n\}$  is *linearly independent* if  $a_1u_1 + \dots + a_nu_n = 0 \Rightarrow a_i = 0, \forall i$ .
- ▶  $S$  is independent  $\Leftrightarrow u$  is independent of  $S \setminus \{u\}$ ,  $\forall u \in S$ .
- ▶  $S$  is *maximally independent* if  $S$  is independent and  $S + \{u\}$  is NOT independent, for each  $u \notin S$ .
- ▶ **Defn** : A *basis* of  $V$  is any independent set  $S$  generating all of  $V$ .
- ▶ Every basis is a maximal independent set of  $V$ .
- ▶ **Theorem** : Every finite  $S$  generating  $V$  has a subset  $T$  which is a basis of  $V$ .
- ▶ **Theorem** : If  $V = \text{span}(S)$  for some finite  $S$ , then every two bases of  $V$  have the same sizes.
- ▶ **Defn** : If  $V = \text{span}(S)$  for some finite  $S$ , the dimension of  $V$ ,  $\dim(V)$ , is the size of any basis of  $V$ .
- ▶ **Example** :  $\dim(R^m) = m$  since  $\{e_1, \dots, e_m\}$  is a basis of  $R^m$ .
- ▶ **Defn** :  $e_j = (e_{ij})_i$  where  $e_{ij} = 1$  if  $i = j$  and  $= 0$  if  $i \neq j$ .

# Bases, coefficient vectors

- ▶ If  $B = (u_1, \dots, u_m)$  is an **ordered** basis of  $V$ , then, for every  $u \in V$ , there exists a *unique* l.c.  $u = \sum_i \alpha_i u_i$  of  $u$ .
- ▶ every ordered basis  $B = (u_1, \dots, u_m)$  gives rise to a bijection  $\phi : V \rightarrow F^m$  mapping  $V$  to coefficient vectors :  $\phi(u) = (\alpha_1(u), \dots, \alpha_m(u))$ .
- ▶ Example of basis to coefficients :
- ▶ If  $\{e_1, e_2\} = \{(1, 0), (0, 1)\}$  and  $\{f_1, f_2\} = \{(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (\frac{1}{2}, \frac{\sqrt{3}}{2})\}$ , then
- ▶  $(x, y) = xe_1 + ye_2$  and  $(x, y) = \frac{(\sqrt{3}x-y)}{\sqrt{3}-1} \cdot f_1 + \frac{2(y-x)}{\sqrt{3}-1} \cdot f_2$ .
- ▶ **Change of Basis :**
- ▶ If  $B = (u_1, \dots, u_n)$  and  $C = (v_1, \dots, v_n)$  are two ordered bases of  $V$ , then there is an invertible  $P \in F^{n \times n}$  :
- ▶ for every  $u \in V$ , its coefficient vectors  $(\alpha_i)_i$  (w.r.t.  $B$ ) and  $(\beta_i)_i$  (w.r.t.  $C$ ) satisfy
$$P \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}.$$
- ▶ Home Work : Determine  $P$ .
- ▶ **Example :**  $B = \{e_1, e_2\}$  and  $C = \{f_1, f_2\}$ .  $P_{B \rightarrow C} = \begin{bmatrix} \frac{\sqrt{3}}{\sqrt{3}-1} & \frac{-1}{\sqrt{3}-1} \\ \frac{-2}{\sqrt{3}-1} & \frac{2}{\sqrt{3}-1} \end{bmatrix}.$

# Isomorphisms

- ▶ **Defn :** An *isomorphism* is a bijective (one-to-one and onto) mapping  $\phi : V \rightarrow W$  such that  $\phi(u + v) = \phi(u) + \phi(v)$  and  $\phi(\alpha u) = \alpha \phi(u)$  for every  $u, v \in V$  and  $\alpha \in F$ .
- ▶
- ▶ **Theorem :** (i) If  $V$  is of dimension  $m$ , then  $V$  and  $F^m$  are isomorphic.
- ▶ (ii) any two  $n$ -dimensional VS's over  $F$  are isomorphic to each other.
- ▶ (iii)  $V$  and  $W$  are isomorphic  $\Rightarrow \dim(V) = \dim(W)$ .
- ▶ **Infer :** it suffices to focus on  $F^m$  to study  $m$ -dimensional spaces over  $F$ .
- ▶
- ▶ **Qn :** Given  $S \subseteq V = R^m$ ,  $|S| = n$ , how do you test if  $S$  is independent ?
- ▶ **Ans :** If  $m < n$ ,  $S$  is *NOT* independent.
- ▶  $m \geq n$  : Form  $A = [u_1, \dots, u_n] \in R^{m \times n}$ , apply GE to  $A$  to obtain  $A'$ . If each of the  $n$  diagonal entries  $\{a'_{11}, \dots, a'_{nn}\}$  is non-zero, then  $S$  is independent, else  $S$  is *NOT* independent.

# Linear Transformations

- ▶ **Defn :** A *Linear Transformation* (LT) is a mapping  $T : V \rightarrow W$  such that  $T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$  for every  $u, v \in V$  and  $\alpha, \beta \in F$ .
- ▶ **Claim :** A LT  $T : V \rightarrow W$  is an isomorphism *if and only if*  $T$  is bijective.
- ▶ A LT is determined completely by the images  $\{T(v_j)\}_j$  of some basis  $\{v_j\}_j$ .
- ▶
- ▶ **Theorem :**  $\dim(V) = n, \dim(W) = m$ . Fix two ordered bases  $B_V = (v_1, \dots, v_n)$  and  $B_W = (w_1, \dots, w_m)$  for  $V$  and  $W$ . There is a bijective function  $\phi$  mapping  $LT(V, W)$  (the set of all LTs mapping  $V$  into  $W$ ) onto  $F^{m \times n}$ .
- ▶
- ▶  $\phi(T) = A$  specified by  $A = (a_{ij})_{ij}$  where the  $j$ -th column  $(a_{ij})_i$  of  $A$  is the coefficient vector of  $T(v_j)$  with respect to ordered  $B_W$ .
- ▶
- ▶ If  $B_V$  is changed to  $C_V$  and  $B_W$  is changed to  $C_W$ , then  $A$  gets changed to  $Q^{-1} \cdot A \cdot P$  where  $P$  and  $Q$  respectively are matrices that characterize the changes  $C_V \rightarrow B_V$  and  $C_W \rightarrow B_W$ .

# Null space, Rank space

- ▶ **Defn** : For any LT  $T$  mapping  $V$  into  $W$ , its
- ▶ *Null-space*,  $N(T)$ , is the subspace  $\{v \in V : T(v) = 0\}$  of  $V$ .
- ▶ *Range-space*,  $R(T)$ , is the subspace  $\{T(u) : u \in V\}$  of  $W$ .
- ▶ *Claim* :  $R(T) = \text{span}(\{T(u_1), \dots, T(u_n)\})$ ,  $\{u_1, \dots, u_n\}$  is any basis of  $V$ .
- ▶
- ▶ **Defn** : For any LT  $T$  from  $V$  into  $W$ , its
- ▶ *Nullity or Null-Rank*,  $n(T)$ , is defined as  $\dim(N(T))$ .
- ▶ *rank*,  $r(T)$ , is defined as  $\dim(R(T))$ .
- ▶
- ▶ **Rank-Nullity Theorem** : For any LT  $T : V \rightarrow W$ ,  $n(T) + r(T) = \dim(V)$ .
- ▶
- ▶ *Claim* : If  $T = A$ ,  $A \in F^{m \times n}$ , wrt ordered  $B_V$  of  $V$  and  $B_W$  of  $W$  respectively, then  $r(T) =$  maximum no of independent columns in  $A = \text{column-rank}(A)$ .

# Linear Transformations - Examples

► Consider  $T_1 : R^3 \rightarrow R^2$  defined by  $T_1(u) = Au$  where  $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ .

►  $N(T_1) = \{u = (x, y, z) : Au = 0\} = \{(a, -a, a) : a \in R\}$  ;  $n(T_1) = 1$ .

►  $R(T_1) = \{Au : u \in R^3\} = R^2$  ;  $r(T_2) = 2$ .  $n(T_1) + r(T_1) = 3$ .



► Consider  $T_2 : R^3 \rightarrow R^4$  defined by  $T_2(u) = Au$  where  $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ .

►  $N(T_2) = \{u = (x, y, z) : Au = 0\} = \{(0, 0, 0)\}$  ;  $n(T_2) = 0$ .

►  $R(T_2) = \text{span}(\{1, 0, 1, 1\}, \{1, 1, 0, 1\}, \{0, 1, 1, 1\})$  ;  $r(T_2) = 3$ .



► Consider  $T_3 : R^3 \rightarrow R^3$  defined by  $T_3(u) = Au$  where  $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ .

►  $N(T_3) = \{u = (x, y, z) : Au = 0\} = \{(0, 0, 0)\}$  ;  $n(T_3) = 0$ .

►  $R(T_3) = \{Au : u \in R^3\} = R^3$  ;  $r(T_3) = 3$ .  $n(T_3) + r(T_3) = 3$ .

# Invertible LTs

- ▶  $T : V_n \rightarrow V_n$  is a LT.  $n$  represents the dimension of  $V$ .
- ▶  $T$  is *invertible* or *non-singular* if  $R(T) = V_n$ .
- ▶ The following are equivalent.
  - ▶  $T$  is invertible.
  - ▶  $N(T) = \{0\} \Leftrightarrow n(T) = 0, r(T) = n$ .
  - ▶  $r(X) = n$  for any  $X \in F^{n \times n}$  representing  $T$  with respect to a basis  $B$  of  $V_n$
  - ▶  $T$  is an isomorphism.
  - ▶  $\exists$  a LT  $S : V_n \rightarrow V_n$  satisfying  $ST = TS = I$  where  $I$  is the identity LT.
- ▶ If  $X$  and  $Y$  represent  $T$  and  $T^{-1}$  w.r.t.  $B$ , then  $Y = X^{-1}$ .

# Algebra of LTs

- ▶  $V_n, V_m$  are VSs over  $F$  of respective dimensions  $n$  and  $m$ .
- ▶  $H(V_n, V_m) := \{f : V_n \rightarrow V_m : f \text{ is a LT}\}$  is a VS over  $F$  of dimension  $nm$  with  $(f + g)(u) = f(u) + g(u)$  and  $(\alpha f)(u) = \alpha f(u)$  for all  $f, u$  and  $\alpha$ .
- ▶  $H(V_n, V_n)$  is a VS of dimension  $n^2$  and  $H(V_n, F)$  is a VS of dimension  $n$ .
- ▶  $H(V_n, V_n)$  is also denoted by  $A_F(V_n)$ .  $H(V_n, F)$  is known as the **dual space** of  $V_n$ .
- ▶ For  $f, g \in H(V_n, V_n)$  define  $f \circ g \in H(V_n, V_n)$  by  $(f \circ g)(u) = f(g(u))$ , for all  $u \in V_n$ .
- ▶
- ▶ An **algebra**  $V$  over  $F$  is a tuple  $(V, F, +, \times, \cdot)$  where
  - (i)  $(V, F, +, \cdot)$  is a vector space over  $F$ .
  - (ii)  $(V, +, \times)$  is an associative ring.
  - (iii)  $\alpha \cdot (u \times v) = (\alpha \cdot u) \times v = u \times (\alpha \cdot v), \forall u, v \in V, \alpha \in F$ .
- ▶ **Claim** : If  $A$  is algebra of dimension  $s$  over  $F$ , then every  $u \in V$  satisfies some polynomial in  $F[x]$  of degree at most  $s$ . Reason : For each  $u, u^s = \alpha_0 1 + \alpha_1 u + \dots + \alpha_{s-1} u^{s-1}$ , for some  $\alpha_i \in F$ .
- ▶
- ▶  $A_F(V_n)$  is an algebra over  $F$  with  $f \times g = f \circ g$ , with identity LT  $I$  being the multiplicative unit element.
- ▶  $\{M : M \in F^{n \times n}\}$  is an algebra over  $F$ , with  $I_n$  as the multiplicative unit.
- ▶ **Theorem** : Every  $M \in F^{n \times n}$  satisfies some polynomial in  $F[x]$  of degree at most  $n^2$ .

# Matrices

- ▶ matrices are also *linear transformations or linear mappings*.
- ▶  $A \in F^{m \times n}$ ,  $x \in F^n$  implies  $Ax \in F^m$ .
- ▶ Linearity :  $A(\alpha x + \beta y) = \alpha(Ax) + \beta(Ay)$ ,  $\forall \alpha, \beta \in F$ .
- ▶ Solving  $Ax = b$  is same as determining a  $x$  which is transformed by  $A$  into  $b$ ,
- ▶ OR can  $b$  be expressed as a *linear combination*  $\sum_{i=1}^n x_i C_i$  of the  $n$  columns of  $A$ .
- ▶ The column-space  $CS(A)$  is the set of *all* linear combinations of columns of  $A$ .
- ▶ The row-space  $RS(A)$  is the set of *all* linear combinations of rows of  $A$ .
- ▶  $rr(A)$ , row-rank of  $A$ , is the maximum number of independent rows of  $A$ .
- ▶  $cr(A)$ , column-rank of  $A$ , is the maximum number of independent columns of  $A$ .
- ▶ **C1**: For any matrix  $A$ ,  $rr(A) = cr(A) = r(A)$ ,  $r(A)$  is the rank of  $A$ .
- ▶  $Ax = b$  has a solution  $\Leftrightarrow cr(A) = cr(A_b)$  where  $A_b = [A, b]$ .
- ▶ Solving  $Ax = b$  reduces to applying a sequence of row operations on  $[A, b]$ ,
- ▶ leading to efficient algorithms for solving systems of linear equations.
- ▶ Gaussian Elimination is one such efficient algorithm.
- ▶

# Gaussian Elimination (outline)

- ▶ Solve for  $x$  satisfying  $Ax = b$  ;
- ▶ **Preprocessing** : If, for some  $i$ ,  $a_{ij} = 0, \forall j$  and  $b_i \neq 0$ , output NO SOLUTION.
- ▶ **Notations** : Augmented matrix  $A_b = [A, b]$ .  $C_1, \dots, C_{n+1}$  - columns of  $A_b$ .  $R_1, \dots, R_m$  - rows of  $A_b$ .
- ▶ **Row operations** :
  - ▶ **RO1** :  $R_i \leftarrow \alpha R_i$  for any  $i$  and any scalar  $\alpha \neq 0$  ;
  - ▶ **RO2** :  $R_i \leftarrow R_i + \alpha R_j$  for any  $i, j$  and scalar  $\alpha \neq 0$  ;
  - ▶ **RO3** : Exchange Rows  $i$  and  $j$  for any  $i \neq j$  ;
- ▶ Find and perform a *suitable* sequence of the above three operations to reduce  $[A, b]$  into  $[A', b']$  where  $A'$  is in row-echelon form.
- ▶ Matrices in row-echelon form look as follows :
$$\begin{bmatrix} 1 & x & 0 & x & 0 & 0 & \dots & x \\ 0 & 0 & 1 & .x & 0 & 0 & \dots & x \\ 0 & .0 & 0 & 0 & 1 & 0 & \dots & x \\ 0 & 0 & 0 & 0 & 0 & 1 & \dots & x \\ 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$
- ▶ Leftmost non-zero element is the *pivot* of a row. Each pivot is 1.
- ▶ Pivot of each row appears to the right of the pivot of the preceding top row.
- ▶ All rows with no pivot (that is, consisting of only 0s) appear at the bottom.

# Gaussian Elimination - continued.

- ▶ Suppose that  $[A, b]$  reduces to  $[A', b']$ ,  $A'$  is in row-echelon form.
- ▶ **C2:**  $RS(A) = RS(A')$ ,  $RS([A, b]) = RS([A', b'])$ .
- ▶ **C3:**  $rr(A) = rr(A')$ ,  $rr([A, b]) = rr([A', b'])$ .
- ▶ **C4:** Each row operation does not change the set of solutions. Hence,  $[A, b]$  admits a solution if and only if  $[A', b']$  admits a solution.
- ▶ **C5:**  $rr(A')$  equals the number of non-zero rows in  $A'$ .
- ▶
- ▶ Gaussian Elimination (GE) : a well-defined procedure (which repeatedly finds a pivot element) to convert  $[A, b]$  into  $[A', b']$  with  $A'$  in row-echelon form.
- ▶ Once  $[A', b']$  is computed, apply Pre-processing to  $[A', b']$  to determine if  $[A', b']$  and hence  $[A, b]$  is *inconsistent* (that is, has no solutions).
- ▶ If  $[A, b]$  is consistent, ignore the bottom  $m - r$  zero rows and focus only on the first  $r$  non-zero rows of  $A'$ .
- ▶ The pivotal columns (those containing pivotal elements) form an upper triangular  $r \times r$  matrix.
- ▶ Choose arbitrary values for variables corresponding to non-pivotal columns.
- ▶ For pivotal columns, get their unique solution conditioned by the choice of values to other variables.
- ▶

# Gaussian Elimination - finding a pivot.

- ▶ Suppose  $A_i$  is the current matrix after  $i$  pivots have been obtained. Let the  $i$  pivots appear in positions  $(1, 1), (2, j_2), \dots, (i, j_i)$  of  $A_i$ .
- ▶ Identify the left-most non-zero column  $s$  in the submatrix  $B$  of  $A_i$  formed by columns numbered  $[j_i + 1..n]$  and rows numbered  $[i + 1..m]$ .
- ▶ Identify the top-most non-zero entry  $x$  in location  $(r, s)$  in  $s$ -th column  $B$ .
- ▶ Apply  $RO_3$  by exchanging  $R_{i+1}$  and  $R_r$ .
- ▶ Apply  $RO_1$  defined by  $R_{i+1} \leftarrow R_{i+1}/x$ .
- ▶ Apply  $RO_2$  for each of the rows numbered  $i + 2$  through  $m$  to make every entry below  $(r, s)$  in the  $s$ -th column a zero.
- ▶ Continue with finding next pivot (if it exists).
- ▶
- ▶ Finding a pivot in  $O(mn)$  elementary arithmetic operations.
- ▶ At most  $r \leq \min\{m, n\}$  pivot elements need to be determined, where  $r = r(A)$ .
- ▶ time complexity  $O(rmn)$  elementary operations.
- ▶

# Gaussian Elimination

- ▶ Example :
- ▶ **E1**  $3x_1 + 4x_2 + 5x_3 + 6x_4 = 3$  ;
- ▶ **E2**  $4x_1 + 5x_2 + 6x_3 + 3x_4 = 4$  ;  $E2 \leftarrow E2 - \frac{4E1}{3}$  ;  $E2 \leftarrow -3E2$  ;
- ▶ **E3**  $5x_1 + 6x_2 + 3x_3 + 4x_4 = 5$  ;  $E3 \leftarrow E3 - \frac{5E1}{3}$  ;  $E3 \leftarrow -3E3/2$  ;
- ▶
- ▶  $E1+E2+E3$  is equivalent to the following  $F1+F2+F3$ .
- ▶ **F1**  $3x_1 + 4x_2 + 5x_3 + 6x_4 = 3$  ;
- ▶ **F2**  $x_2 + 2x_3 + 15x_4 = 0$  ;
- ▶ **F3**  $x_2 + 8x_3 + 9x_4 = 0$  ;  $E3 \leftarrow E3 - E2$  ;  $E3 \leftarrow -E3/6$  ;
- ▶
- ▶  $F1 + F2 + F3$  is equivalent to the following  $G1 + G2 + G3$ .
- ▶ **G1**  $3x_1 + 4x_2 + 5x_3 + 6x_4 = 3$  ;
- ▶ **G2**  $x_2 + 2x_3 + 15x_4 = 0$  ;
- ▶ **G3**  $x_3 - x_4 = 0$  ;
- ▶ Thus,  $x_3 = x_4$  ;  $x_2 = -17x_4$  ;  $x_1 = 1 + 19x_4$ .
- ▶ Special structure of  $G1 + G2 + G3$  enables us to obtain  $x_1, x_2, x_3$  in terms of  $x_4$ .
- ▶ system of equations is said to be in row-reduced echelon form.

# Gaussian Elimination - Time Complexity

- ▶ Row-echelon form matrices :

- ▶ 
$$\begin{bmatrix} 1 & x & 0 & x & 0 & 0 & \dots & x \\ 0 & 0 & 1 & .x & 0 & 0 & \dots & x \\ 0 & .0 & 0 & 0 & 1 & 0 & \dots & x \\ 0 & 0 & 0 & 0 & 0 & 1 & \dots & x \\ 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

- ▶
- ▶ Leftmost non-zero element is the *pivot* element of a row.
- ▶ Finding a pivot in  $O(mn)$  elementary arithmetic operations.
- ▶ At most  $m$  pivot elements need to be determined.
- ▶ time complexity  $O(m^2n)$  elementary operations.
- ▶ It is  $O(n^3)$  operations if  $m \leq n$ .
- ▶  $m > n$  ? One needs to find only at most  $\min\{m, n\}$  pivots.
- ▶ When  $m > n$ , it takes  $O(mn^2)$  operations.

# Gaussian Elimination - Solving several systems

- ▶  $F$  is a field like, say,  $\mathbb{Q}$  or  $\mathbb{R}$ .
- ▶ For  $A \in F^{m \times n}$  and  $b_1, \dots, b_k \in F^m$ ,
- ▶ Find  $x_1, \dots, x_k \in F^n$  satisfying  $Ax_j = b_j$  for  $j = 1, \dots, k$ .
- ▶ Perform Gaussian Elimination on  $[A, b_1, \dots, b_k]$  to obtain  $[A', b'_1, \dots, b'_k]$ .  $A'$  is in row-echelon form.
- ▶ Recover, for each  $j$ , the solution from  $A'x_j = b'_j$ .
- ▶ Operations complexity =  $O(m^2(n+k))$  or  $O(mn(n+k))$ .

# Gaussian Elimination - Computing the inverse

- ▶ Multiplicative inverse of  $A$ ,  $A^{-1}$ , is any  $X \in F^{n \times n}$  satisfying  $XA = AX = I_n$ .
- ▶ Inverse, if it exists, is unique.
- ▶ Write  $I_n = (e_1, \dots, e_n)$  where  $e_j$  has 0 in every component except the  $j$ -th component where it is 1.
- ▶ Write  $X = (x_1, \dots, x_n)$ .
- ▶  $AX = I_n$  is same as  $(Ax_1, \dots, Ax_n) = (e_1, \dots, e_n)$  which is same as  $Ax_j = e_j$  for each  $j = 1, \dots, n$ .
- ▶ Perform GE on  $[A, e_1, \dots, e_n]$  to obtain  $[A', e'_1, \dots, e'_n]$ .
- ▶ If the modified system is solvable (can be verified easily), then  $X$  exists and  $x_j$  is the solution of  $A'x_j = e'_j$  which is easily recoverable.

# Computing the Inverse through GE-Example

- ▶ Let  $A$  be defined by  $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 2 \\ 4 & 2 & 3 \end{bmatrix}$ .
- ▶ GE applied to  $[A, e_1, e_2, e_3]$  gives  $[A', e'_1, e'_2, e'_3]$  by :
- ▶  $\begin{bmatrix} 2 & 3 & 4 & 1 & 0 & 0 \\ 3 & 4 & 2 & 0 & 1 & 0 \\ 4 & 2 & 3 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -4 & -\frac{3}{2} & 1 & 0 \\ 0 & -4 & -5 & -2 & 0 & 1 \end{bmatrix} \rightarrow$
- ▶  $\begin{bmatrix} 2 & 3 & 4 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -4 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 27 & 10 & -8 & 1 \end{bmatrix}$ .
- ▶ Solving for  $x_j$  in  $A'x_j = e'_j$ , we obtain  $X = A^{-1}$  as
- ▶  $X = \frac{1}{27} \cdot \begin{bmatrix} -8 & 1 & 10 \\ 1 & 10 & -8 \\ 10 & -8 & 1 \end{bmatrix}$  ;

# Computing the Inverse through GE-Example

- ▶ Let  $A$  be defined by  $A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 3 \end{bmatrix}$ .

- ▶ GE applied to  $[A]$  gives  $[A']$  which is

- ▶  $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

- ▶ Ops :  $R4 \leftarrow R4 - R1$  ;  $R4 \leftarrow R4 - R2$  ;  $R4 \leftarrow R4 - R3$  ;
- ▶ The last row of  $A'$  implies that  $A'$  and hence  $A$  are not invertible. Why ?

# Determining a basis of $R(T)$

- **Qn** : Given  $A \in F^{m \times n}$  : Determine a basis for  $R(A) = \text{span of columns of } A$ .
- **Soln** : Apply GE to  $A$  to get  $A'$ . Basis of  $R(A)$  = the columns of  $A$  corresponding to those in  $A'$  containing a pivot element.

- Example : determine a basis of  $R(A)$  where  $A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ .2 & -1 & -4 & 8 \\ . -1 & 1 & 3 & -5 \\ -1 & 2 & 5 & -6 \\ -1 & -2 & -3 & 1 \end{bmatrix}$ .

- Applying GE to  $A$ , we obtain

- $\begin{bmatrix} 1 & 2 & 3 & -1 \\ .0 & -5 & -10 & 10 \\ .0 & 3 & 6 & -6 \\ 0 & 4 & 8 & -7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & -1 \\ .0 & -5 & -10 & 10 \\ .0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & -1 \\ .0 & -5 & -10 & 10 \\ .0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ .

- Columns  $C_1$ ,  $C_2$  and  $C_4$  of  $A$  form a basis for  $R(A)$ .

# row-rank = column-rank

- ▶ **Definitions :**

- ▶  $A \in F^{m \times n}$  arbitrary.
- ▶  $\text{row-space}(A)$  = span of rows of  $A$ , is a subspace of  $F^n$ .
- ▶  $\text{column-space}(A)$  = span of columns of  $A$ , is a subspace of  $F^m$ .
- ▶  $\text{row-rank}(A)$  = dimension of  $\text{row-space}(A)$ , shortly  $rr(A)$ .
- ▶  $\text{column-rank}(A)$  = dimension of  $\text{column-space}(A)$ , shortly  $cr(A)$ .

▶

- ▶ **Theorem :** For any  $A$ ,  $\text{row-rank}(A) = \text{column-rank}(A)$ .

▶

- ▶ **Defns :** Rank of  $A$ , shortly  $rk(A)$ , equals the common value.
- ▶  $A$  has *full-rank* if  $rk(A) = \min\{m, n\}$ . That is, either  $A$  has full column-rank  $n$  or full row-rank  $m$ .

▶

- ▶ **Claims :**

- ▶ For  $b \in F^{m \times n}$ ,  $Ax = b$  admits a solution  $\Leftrightarrow cr([A, b]) = cr(A)$ .
- ▶  $A \in F^{n \times n}$  is invertible  $\Leftrightarrow rk(A) = n$ .

# Solving $Ax = b$

- ▶  $A \in R^{m \times n}$  ;  $A$  has full-rank.
- ▶ **Case**  $m = n$  :  $Ax = b$  has a unique solution  $x = A^{-1}b$ .
- ▶ **Case**  $m < n$  :  $Ax = b$  either does not have any solution or has infinitely many solutions.
- ▶ **Case**  $m > n$  :  $Ax = b$  either has a unique solution or has no solution.
- ▶
- ▶  $A \in R^{m \times n}$  :  $A$  does not have full rank.
- ▶  $Ax = b$  has either no solution or has infinitely many solutions.
- ▶
- ▶  $N(A)$  = the set of solutions of the homogenous system  $Ax = 0$ .
- ▶ Suppose  $Ax = b$  is solvable ; Let  $x^* \in R^n$  be a specific solution of  $Ax^* = b$ .
- ▶ For each  $b \in R^m$ , Solution Space of  $(Ax = b) = \{x^* + z : z \in N(A)\}$ .
- ▶ It is either the emptyset or an affine subspace of dimension  $n(A)$ .

# Row-equivalent matrices

- ▶  $O1 (R_i \leftarrow \alpha R_i)$  is the same as  $A \leftarrow I_m^{i,\alpha} \cdot A$  for  $\alpha \neq 0$ ;
- ▶  $I_m^{i,\alpha}$  is  $I_m$  with  $i$ -th row multiplied by  $\alpha$ .  $I_m^{i,\alpha}$  is invertible for  $\alpha \neq 0$ .
- ▶ Example :  $I_4^{2,0.5} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ .
- ▶
- ▶  $O2 (R_i \leftarrow R_i + \alpha R_j)$  is the same as  $A \leftarrow (I_m + \alpha Z_m^{i,j}) \cdot A$  ;
- ▶  $Z_m^{i,j}$  is the  $m \times m$  zero matrix  $Z_m$  except  $(Z_m^{i,j})_{ij} = 1$ .  $I_m + \alpha Z_m^{i,j}$  is invertible for  $i \neq j$  and any  $\alpha$ .
- ▶ Example :  $Z_4^{2,3} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ .
- ▶

# Row-equivalent matrices

- ▶  $O3 (R_i \leftrightarrow R_j)$  is the same as  $A \leftarrow I_m^{i,j} \cdot A$  ;
- ▶  $I_m^{i,j}$  is the identity matrix  $I_m$  except that Rows  $i$  and  $j$  are interchanged.  $I_m^{i,j}$  is invertible.
- ▶ Example :  $I_4^{2,3} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ .
- ▶
- ▶ Each of  $\{O1, O2, O3\}$  is equivalent to pre-multiplying by an invertible elementary matrix.
- ▶ An **elementary** matrix is obtained from  $I_m$  by performing one of  $\{O1, O2, O3\}$ .
- ▶ **Theorem** : A sequence  $O_{i_1}, \dots, O_{i_r}$  of row-operations performed on  $A$  is equivalent to performing  $A \leftarrow E_r \cdot E_{r-1} \dots E_1 \cdot A$ .  $E_i$ 's are appropriate invertible elementary matrices.

# Row-Equivalent matrices

- ▶  $A, B \in F^{m \times n}$ ;  $A$  and  $B$  are **row equivalent (RE)** if  $\exists$  a sequence of row operations converting  $A$  into  $B$ .
- ▶ **Theorem** : Below,  $A, B \in F^{m \times n}$  and  $b \in F^m$  are arbitrary :
- ▶ (a) Row-equivalence is an equivalence relation on  $F^{m \times n}$ .
- ▶ (b)  $A$  and  $B$  are **RE** if and only if  $B = PA$  for some invertible  $P \in F^{m \times m}$ .
- ▶ (c)  $A$  and  $B$  are **RE**  $\Rightarrow r(A) = r(B)$ .
- ▶ (c1)  $r(A) = k \iff A$  is RE to a matrix for the form  $B = \begin{bmatrix} A \\ Z \end{bmatrix}$ , where  $A \in F^{k \times n}$  and  $Z \in F^{(m-k) \times n}$  and  $A$  is in row-echelon form.
- ▶ (d)  $A \in F^{n \times n}$  is non-singular (invertible)  $\iff A$  and  $I_n$  are **RE**.
- ▶ (d')  $A \in F^{n \times n}$  is non-singular  $\iff A =$  product of elementary matrices.
- ▶ (e)  $A, B$  are in reduced row-echelon form and are **RE**  $\Rightarrow A = B$ .
- ▶ (f) Every  $A$  is **RE** to a unique  $C$  in reduced row-echelon form.
- ▶ (g) the systems  $Ax = b$  and  $Bx = b$  have the same solutions  $\iff A' = B'$  where  $A'$  and  $B'$  are their unique reduced row-echelon representations.
- ▶ (h)  $A$  and  $B$  are RE  $\Rightarrow A$  and  $B$  represent the same LT  $T : F^n \rightarrow F^m$ , except for the bases used for  $F^m$ .
- ▶

# Row-Equivalent matrices

- ▶ **Corollary** : The following are equivalent for any  $A \in F^{n \times n}$ ,  $b \in F^n$  :
- ▶ (a)  $A$  is invertible (non-singular).
- ▶ (b)  $rk(A) = n$
- ▶ (c) rows of  $A$  are independent.
- ▶ (d) columns of  $A$  are independent.
- ▶ (e)  $A$  and  $I_n$  are **RE**.
- ▶ (f)  $A$  is a product of elementary matrices.
- ▶ (g)  $Ax = b$  has a unique solution, namely,  $x = A^{-1}b$ .

# Column-equivalent matrices

- ▶  $O1'$  ( $C_i \leftarrow \alpha C_i$ ) is the same as  $A \leftarrow A \cdot I_n^{i,\alpha}$  for  $\alpha \neq 0$ ;
- ▶  $I_n^{i,\alpha}$  is  $I_n$  with  $i$ -th column multiplied by  $\alpha$ .  $I_n^{i,\alpha}$  is invertible for  $\alpha \neq 0$ .
- ▶ Example :  $I_4^{3,0.5} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ .
- ▶
- ▶  $O2'$  ( $C_i \leftarrow C_i + \alpha C_j$ ) is the same as  $A \leftarrow A \cdot (I_n + \alpha Z_n^{j,i})$  ;
- ▶  $Z_n^{j,i}$  is the  $n \times n$  zero matrix  $Z_n$  except  $(Z_n^{j,i})_{ji} = 1$ .  $I_n + \alpha Z_n^{j,i}$  is invertible for  $i \neq j$  and any  $\alpha$ .
- ▶ Example :  $Z_4^{2,3} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ .
- ▶

# Column-equivalent matrices

- ▶  $O3' (C_i \leftrightarrow C_j)$  is the same as  $A \leftarrow A \cdot I_n^{i,j}$  ;
- ▶  $I_n^{i,j}$  is the identity matrix  $I_n$  except that Columns  $i$  and  $j$  are interchanged.  $I_n^{i,j}$  is invertible.
- ▶ Example :  $I_4^{2,3} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ .
- ▶
- ▶ Each of  $\{O1', O2', O3'\}$  is equivalent to post-multiplying by an invertible elementary matrix.
- ▶ **Theorem** : A sequence  $O_{i_1}', \dots, O_{i_r}'$  of column-operations performed on  $A$  is equivalent to performing  $A \leftarrow A \cdot E_1 \cdot E_2 \dots E_r$ .
- ▶  $A$  and  $B$  are column equivalent (**CE**) if  $B = AP$  for some non-singular  $P \in F^{n \times n}$ . Being **CE** is an equivalence relation.
- ▶  $A$  and  $B$  are **CE**  $\Rightarrow rk(A) = rk(B)$ .

# Equivalent matrices

- ▶  $A, B \in F^{m \times n}$  are equivalent (**EQ**)  $B = QAP$  for invertible square matrices  $Q, P$ .
- ▶ Being **EQ** is an equivalence relation over  $m \times n$  matrices.
- ▶ **Thm** : Every  $A \in F^{m \times n}$  with rank  $k$  is equivalent to a unique  $\begin{bmatrix} I_k & Z \\ Z & Z \end{bmatrix}$  - its *canonical form or representation*.
- ▶ **Corollary** : Equivalence class of a matrix is determined by its rank.
- ▶